Beyond Recency: Mechanism-Aware Off-Policy Evaluation in Non-Stationary Environments

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Abstract. Off-policy evaluation for the current period must often rely on logs collected over many earlier periods, with only a small sample from the period itself. The usual remedies for non-stationarity, recency windows, decay and forecasting, treat every change in the logs alike, so they discard data that is still informative. We argue that the useful question is not how recent a period is but which mechanism changed: a change in the context distribution or in the logging policy can be corrected by reweighting, while only a change in the conditional reward biases a reweighted estimate. We propose O-DRIFT, which segments the history with a per-action residual test that is invariant to context and logger drift, fits joint reward models across segments, reuses older segments with a drift correction, and pools the segment estimates at minimum variance with a random-effects interval. In a pre-registered evaluation with a one-shot test split, O-DRIFT outperforms a cross-domain estimator that groups periods by mean reward in all 15 reward-drift settings, and replacing its residual test by mean-reward grouping causes spurious splits under context and logger drift. On semi-synthetic Open Bandit data, however, drift-corrected reuse is strongly biased; we report this failure and show why validation on the small current-period sample cannot repair it.

Keywords: Off-policy evaluation · Non-stationarity · Concept drift · Recommender systems

1 Introduction

Recommender and advertising systems decide whether to deploy a new policy by estimating its value from logged interactions of the policies that ran before it. Off-policy evaluation (OPE) makes this possible when the logs record the propensities of the logging policy [10,8]. In practice the logs span many weeks, and the environment does not stand still: users change, the deployed logger is replaced, and the click-through behaviour of the same users on the same items moves. What a team needs is the value of the candidate policy *now*, in a target period for which only a small sample has been collected.

Two simple answers bracket the problem. Pooling all history gives a low-variance estimate that is biased whenever the environment has changed. Using only the small target sample (the *anchor*) is unbiased but noisy. Existing

remedies interpolate between the two along the time axis: sliding windows and moving averages [11], exponential decay, forecasting the policy value as a time series [20,3], and regression-assisted estimators that reuse a model fitted on recent data [13]. All of them assume that older data is less useful because it is older.

This assumption ignores what changed. A logged record is generated by three mechanisms: the context distribution $P_t(x)$, the logging policy $\mu_t(a \mid x)$ and the conditional reward $q_t(x, a) = \mathbb{E}[R \mid x, a, t]$. The first two are nuisances that standard tools already correct: importance weights handle the logger, and density-ratio transport handles the contexts [21,1]. Only a change in q_t biases a correctly reweighted estimate. An old period whose users looked different but clicked in the same way is as informative as yesterday; a recent period after a change in click behaviour is not. Recency is a proxy for the mechanism that matters, and a poor one when contexts or loggers drift more often than rewards.

We turn this observation into an estimator, O-DRIFT (OPE under DRIFT). It tests for drift in the conditional reward directly, with a per-action residual statistic computed from one pooled reward model, and segments the history by binary segmentation. Under correct specification the test keeps its nominal level under any context or logger drift (Proposition 1), whereas statistics built from raw rewards, such as the mean-reward similarity used by the cross-domain estimator COPE [14], do not. Within each segment O-DRIFT pools windows with balanced multiple-logger weights; across segments it fits joint reward models with a shared base and per-segment offsets, reuses older segments after correcting their reward offset, and combines the segment estimates with the anchor at minimum variance.

Our contributions are:

- a mechanism decomposition of drift for OPE and a reward-drift monitor whose size does not depend on context or logger drift (Sections 3 and 4);
- the O-DRIFT estimator with segment-wise joint reward models, drift-corrected reuse, minimum-variance pooling and a random-effects interval (Section 4);
- a pre-registered evaluation on simulated regimes and semi-synthetic Open Bandit data [16], with a one-shot test split, a COPE baseline and a grouping ablation that isolates the monitor (Sections 5 and 6).

The evaluation also yields a negative result that we report in full: on Open Bandit, where the reward model is misspecified and drift is large, drift-corrected reuse is heavily biased, and two pre-specified attempts to validate reuse on the anchor or on a held-out window failed, because those samples are too small to detect the bias that matters (Section 6).

2 Related Work

Off-policy evaluation. Inverse propensity scoring [10] and the doubly robust (DR) estimator [8] are the standard tools for contextual bandit OPE; we cross-fit the reward models as in double machine learning [5]. When the data come

Table 1. What existing estimators adapt to. Context and logger drift are handled by reweighting; "reward drift" means the method decides which data to reuse according to changes in $q_t(x, a)$.

Method	Context drift	Logger drift	Reward drift	Uses anchor	Interval
Sliding window / decay [11]	–	✓	by recency	–	–
Forecasting [20,3]	–	✓	by trend	–	✓
OPFV [17]	–	✓	by time features	–	–
Regression-assisted DR [13]	–	✓	by recency	✓	✓
SLOPE / MAGIC [18,19]	–	✓	by bias–variance	✓	–
COPE [14]	–	✓	by mean reward	✓	–
O-Drift (ours)	✓	✓	by residual test on q_t	✓	✓

from several loggers, weighting each record by the mixture of the logging policies (balanced or multiple importance weighting) reduces variance and stays unbiased [1,12]. Covariate shift between the logging and target populations is handled by density-ratio transport [21]. O-DRIFT uses all three ingredients; what it adds is the decision of which historical data the reweighted estimate may use.

Non-stationary OPE. Jagerman et al. [11] study sliding-window and exponentially decayed estimators for non-stationary bandits. Thomas et al. [20] and Chandak et al. [3,4] forecast the future value of a policy from its past per-period estimates, and Shimizu et al. [17] estimate future values by exploiting time features shared between past and future. Liu et al. [13] reuse a reward model fitted on older data inside a DR estimator on the new data. These methods weight data by time or by a model of the trend in the value; none distinguishes the mechanism behind a change.

Cross-domain OPE. COPE [14] treats other domains as sources for a target domain, clusters domains by the similarity of their mean rewards and combines multiple importance weighting inside the target cluster with a decomposed reward model. Treating time windows as domains makes COPE the closest existing estimator to ours. Its clustering statistic, the mean reward, mixes all three mechanisms; our monitor isolates the one that biases reweighted estimates. We include COPE as a tuned baseline and as a grouping ablation.

Estimator selection and drift detection. MAGIC [19], SLOPE [18] and adaptive weighting [22] choose among or blend estimators of increasing bias and decreasing variance; we use the first two as baselines over nested recency windows. Drift detection on data streams is surveyed by Gama et al. [9]. Our monitor is a heterogeneity test in the sense of Cochran [6] applied to per-action residuals, and our interval is the DerSimonian–Laird random-effects interval [7]; we claim novelty for the statistic being tested, not for the tests. Choosing segments by a preliminary test is a pre-test estimator [2], and it shares the known risk of such estimators near the detection threshold.

3 Problem Setting

Data. Time is divided into windows $k = 1, \dots, K$ followed by a target period T. Window k contributes logged records $\mathcal{D}_k = \{(x_i, a_i, r_i, \mu_k(a_i \mid x_i))\}_{i=1}^{n_k}$ with contexts $x \sim P_k$, actions $a \sim \mu_k(\cdot \mid x)$ from the logger deployed in window k, and binary rewards with conditional mean $q_k(x, a)$. From the target period we have a small anchor $\mathcal{D}_T$ of $n_T \ll \sum_k n_k$ logged records and a sample $\{x_j^*\}_{j=1}^{M}$ of unlabelled target contexts. The estimand for a target policy π is

$$V_T(\pi) = \mathbb{E}_{x \sim P_T}\Big[\sum_a \pi(a \mid x)\, q_T(x, a)\Big]. \tag{1}$$

Three drift mechanisms. A window can differ from the target in P_k, in μ_k or in q_k. With logged propensities and a density ratio $w_k(x) = P_T(x)/P_k(x)$, the reweighted DR score of a record in window k,

$$\Gamma_k(x, a, r) = \textstyle\sum_{a'} \pi(a' \mid x)\hat{q}(x, a')\big|_{x \sim P_T} + w_k(x)\, \frac{\pi(a|x)}{\mu_k(a|x)}\,\big(r - \hat{q}(x, a)\big), \tag{2}$$

has expectation $\mathbb{E}_{P_T}[\sum_a \pi(a \mid x) q_k(x, a)]$ for any $\hat{q}$. It equals $V_T(\pi)$ exactly when $q_k = q_T$ on the support of π under P_T. Context and logger drift therefore cost variance but not bias; reward drift costs bias that no reweighting removes. The transfer bias of window k is $b_k = \mathbb{E}_{P_T}[\sum_a \pi(a \mid x)(q_k - q_T)(x, a)]$.

Assumptions. (A1) Logged propensities are the ones used at event time. (A2) Overlap: $\pi(a \mid x) > 0$ implies $\mu_k(a \mid x) > 0$ for the windows used, and the context ratio w_k is bounded. (A3) Reward drift is piecewise: the windows split into contiguous segments with a common q within each segment. (A4) For drift-corrected reuse, the difference between segment rewards lies in the reward-model class (on the logit scale). Our regimes violate each assumption in turn (S6 breaks A1, S7 A2, S3b and S5 A3, S8 and the misspecified cells A4) so that we can see where the method fails.

What the anchor can and cannot do. Without an anchor, b_k is not identified from history: any q_T that agrees with nothing in the logs is consistent with them, so no method can certify an estimate. With an anchor, b_k is identified, but only up to the noise of the anchor estimate, whose standard deviation is of order $\sigma/\sqrt{n_T}$. Biases below this level cannot be detected from the anchor, however the test is built. This limit governs the failure we report in Section 6.

4 Method

O-DRIFT has four steps (Algorithm 1): segment the history by reward drift, fit joint reward models, form one drift-corrected estimate per segment, and pool them with the anchor.

4.1 A reward-drift monitor

We fit one pooled per-action logistic model $\tilde{q}$ with linear and quadratic context terms on all historical windows. For a block B of windows and an action a, let $m_B(a)$ be the mean residual $r_i - \tilde{q}(x_i, a)$ over records in B with $a_i = a$, and $v_B(a)$ its binomial variance estimate. Two adjacent blocks L, R are compared by

$$\chi^2_{LR} = \sum_a \frac{\big(m_L(a) - m_R(a)\big)^2}{v_L(a) + v_R(a)}, \tag{3}$$

referred to a χ^2 distribution with one degree of freedom per action with enough records. Binary segmentation splits the history at the most significant boundary while the p-value is below α, giving contiguous segments $S_1, \ldots, S_J$ ordered from newest to oldest.

Proposition 1 (Invariance to context and logger drift). *Suppose $q_k = q$ for all windows in $L \cup R$ and $\tilde{q} \to q$. Then for every action a, $\mathbb{E}[m_L(a) - m_R(a)] \to 0$ whatever the context laws P_k and loggers μ_k, so (3) is asymptotically χ^2 and the test has level α under any context or logger drift. The mean-reward statistic $\bar{r}_L - \bar{r}_R$ and the per-action raw-reward statistic do not have this property.*

Proof (sketch). In window k, $\mathbb{E}[R - q(X, a) \mid A = a, k] = \mathbb{E}_{P_k(x|a)}[q_k(x, a) - q(x, a)] = 0$, because $P_k(x \mid a) \propto P_k(x)\mu_k(a \mid x)$ only reweights a function that is identically zero. Only $q_k \neq q$ moves $m_B(a)$. In contrast, $\mathbb{E}[R \mid k] = \sum_a \int P_k(x)\mu_k(a \mid x)q(x, a)\, dx$ changes with P_k and μ_k when q is fixed, which gives false alarms; and a reward change can be cancelled by an opposite context or logger change, which gives missed drift. The per-action raw mean $\mathbb{E}_{P_k(x|a)}[q(x, a)]$ removes the action mix but still depends on P_k and on μ_k as a function of x.

If q lies outside the model class, the residual has conditional mean $\mathbb{E}_{P_k(x|a)}[q - \tilde{q}]$, which depends on P_k; context drift can then trigger the monitor. The richer the model class, the smaller this effect.

4.2 Joint segment reward models

Reward models for estimation are cross-fitted [5] per-action logistic regressions on a context basis $\phi(x)$. Across segments we fit one joint model

$$\operatorname{logit} \hat{q}_j(x, a) = \phi(x)^\top \beta_a + \phi(x)^\top \delta_{j,a}, \qquad \delta_{1,a} = 0, \tag{4}$$

a shared base plus per-segment offsets, with an L_2 penalty on the offsets scaled by $1/s^2$. The scale $s \leq 1$ acts as a ridge prior on drift size and interpolates between fully corrected ($s = 1$) and uncorrected reuse ($s \to 0$). The basis is chosen per episode between linear features and $D = 32$ random Fourier features [15] by held-out log-loss of the pooled model; the same basis is used by every method.

Algorithm 1 O-DRIFT for one target policy π

Require: windows $\mathcal{D}_1..\mathcal{D}_K$, anchor $\mathcal{D}_T$, target contexts $x^*_{1..M}$, loggers μ_k
1: Fit pooled $\tilde{q}$; segment the history by (3) at level $\alpha \to S_1, \ldots, S_J$
2: Choose the context basis by held-out log-loss; fit joint models $\hat{q}_j$ with offset scale s
3: Choose the reference segment with the anchor test (z_{anc})
4: For each segment compute $\hat{V}_j$ by (5); compute the anchor DR estimate
5: Pool at minimum variance; return $\hat{V}$ and the DerSimonian–Laird interval

4.3 Reference segment and drift-corrected reuse

The reference segment is the newest segment whose uncorrected estimate the anchor does not contradict at $|z| \le z_{\text{anc}}$; if all are contradicted the anchor is used alone. Each segment S_j contributes

$$\hat{V}_j = \frac{1}{M}\sum_{m=1}^{M}\sum_{a}\pi(a \mid x^*_m)\hat{q}_{\text{ref}}(x^*_m, a) + \frac{1}{n_j}\sum_{i\in S_j} W_i\left(r_i - \hat{q}_j(x_i, a_i)\right), \tag{5}$$

$$W_i = \frac{\pi(a_i \mid x_i)}{\sum_{k\in S_j}\frac{n_k}{n_j}\mu_k(a_i \mid x_i)/w_k(x_i)}. \tag{6}$$

The direct-method term uses the reference model, so it targets q_{ref}; the correction term uses the segment's own model, so it removes the segment's own model error. W_i is the balanced importance weight from the target to the segment's mixture of loggers and context laws [1,12], with w_k estimated by a probabilistic classifier. For the reference segment $\hat{q}_j = \hat{q}_{\text{ref}}$ and (5) is the usual balanced DR estimate. For an older segment, the difference $\hat{q}_{\text{ref}} - \hat{q}_j$ is exactly the fitted reward drift, so (5) transfers the segment's evidence to the reference reward when (A4) holds.

4.4 Pooling and interval

The anchor DR estimate joins the segment estimates, and the weights $\hat{\alpha} = \arg\min_{\alpha\ge 0,\, \mathbf{1}^\top\alpha=1}\alpha^\top\hat{\Sigma}\alpha$ minimise the estimated variance, where $\hat{\Sigma}$ accounts for the target contexts shared by all direct-method terms. Because a residual bias may remain, we report the DerSimonian–Laird interval [7], which inflates the standard error by the between-component heterogeneity $\hat{\tau}^2$. As a certified alternative we also report a sample-split interval with an explicit bias allowance; it is valid but wide. Without an anchor O-DRIFT returns an estimate flagged as not certified.

Settings. $\alpha = 10^{-4}$ and $z_{\text{anc}} = 5$ were tuned on the development split before the first test run; $s = 0.25$ was tuned on the development split of the pre-registered revision that we test (Section 5).

5 Experimental Setup

Simulated regimes. Each environment has $d = 5$ Gaussian contexts, 10 actions and per-action logistic rewards with a linear part and a nonlinear part of weight 0.5 ("misspecification 0.5") or 3.0. The level 3.0 was fixed by matching the nonlinear share of logit variance in Open Bandit (0.43 vs. 0.41) before any result at that level was seen. The history has $K = 12$ windows of 5,000 records; the anchor has 1,000 records and there are 10,000 target contexts. Ground truth is exact over actions and Monte Carlo over 10^6 contexts. The regimes isolate one mechanism at a time: S0 stationary; S1 context drift growing with age; S2 logger switches A→B→C; S3a abrupt reward drift in the older half; S3b gradual reward drift towards the target; S4 mixed (oldest third reward-drifted, middle third context-shifted, newest third poorly explored); S5 seasonal context and reward drift; S6 wrong logged propensities (violates A1); S7 recent windows that never play three actions (violates A2); S8 reward drift along an out-of-class interaction, S8b the same without an in-class component (violate A4). Intensities are given with each regime in Table 2. Four candidate policies of increasing noise around the optimal policy are evaluated in every episode.

Open Bandit semi-synthetic data. We use the uniformly random logging data of the Men campaign of the Open Bandit Dataset [16] (452,949 records, 34 items). Contexts are the real user features reduced to 8 principal components, in their real temporal order, split into 13 blocks of equal size (12 history windows and the target block). The base reward is a per-action logistic model with linear and quadratic terms fitted to the real clicks and shifted to a mean click rate of 0.05; actions, propensities, rewards and the injected logger and reward drift are synthetic. Because the generating model is quadratic, the methods' reward models are misspecified. These data test robustness, not natural drift.

Baselines. All estimators share the reward models and the per-episode basis rule. We compare pooled DR; anchor-only DR; sliding-window DR (recent k windows + anchor) and exponentially decayed DR [11]; minimum-variance window pooling; density-ratio transport DR and its recent-k, decayed and balanced-decayed variants; regression-assisted DR [13]; SLOPE [18] and MAGIC [19] over nested recency windows; forecasting of per-window values [20]; a time-aware DR whose reward model includes a polynomial time trend; balanced multiple-logger DR [1]; an anchor-validated window selector; COPE [14] with time windows as domains, the anchor as the target domain and the cluster size tuned; and "monitor-select", DR on the newest monitor segment plus the anchor without reuse.

Protocol. The protocol (`configs/protocol.yaml` in the code) was fixed before tuning. Every method family, baselines included, is tuned on a development split (four environments, 200 episodes per cell) by minimising the mean over cells of log RMSE. Test splits use unseen environments (300 episodes per cell; 200 for Open Bandit) and are run once per frozen configuration. The version reported

Table 2. One-shot test results on the simulated regimes. RMSE ($\times 10^{-3}$) of O-DRIFT, of its configuration without offset shrinkage (v1) and of the best tuned baseline in the cell; ratio of O-DRIFT to that baseline (*: paired 95% interval of the RMSE difference excludes 0); coverage of the nominal 95% heterogeneity-adjusted interval. 300 episodes $\times$ 4 policies per cell.

Regime	Int.	Misspec.	O-DRIFT	v1	Best tuned baseline	RMSE	Ratio	Cov.
S0	0	0.5	5.4	5.4	Min-var DR	5.1	1.05*	0.93
S1	0.5	0.5	5.5	5.5	COPE	5.5	1.00	0.95
S1	2	0.5	8.0	8.0	COPE	7.8	1.02	0.93
S2	0	0.5	5.1	5.1	Balanced DR	4.7	1.07*	0.95
S3a	0.25	0.5	6.4	6.5	Monitor-select DR	7.0	0.93*	0.89
S3a	1	0.5	9.8	6.5	Monitor-select DR	7.0	1.40*	0.83
S3b	0.25	0.5	10.9	10.8	Time-DR	8.5	1.28*	0.63
S3b	1	0.5	26.6	18.4	Time-DR	8.8	3.02*	0.27
S4	0.25	0.5	7.5	8.0	Balanced DR	8.6	0.86*	0.95
S4	1	0.5	14.1	14.8	T-recent DR	21.7	0.65*	0.78
S5	0.5	0.5	21.5	20.9	Decay DR	20.4	1.05*	0.34
S5	1	0.5	34.7	28.9	SLOPE	36.0	0.96	0.22
S6	0	0.5	5.9	5.9	Balanced DR	5.3	1.13*	0.94
S7	0	0.5	7.4	7.4	Min-var DR	4.9	1.52*	0.94
S8	0.5	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.88
S8	1	0.5	6.9	6.8	Monitor-select DR	7.0	0.98	0.86
S8b	1	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.89
S0	0	3	4.9	4.9	Pooled DR	4.5	1.08*	0.95
S2	0	3	5.0	5.0	Balanced DR	4.6	1.08*	0.95
S3a	1	3	6.5	6.7	Monitor-select DR	6.7	0.98	0.86
S4	1	3	17.9	25.7	TB-decay DR	22.0	0.81*	0.70
S5	1	3	23.3	26.2	Monitor-select DR	29.2	0.80*	0.33
S8	1	3	6.6	6.7	Monitor-select DR	6.7	0.99	0.86

here is the third pre-registered one. The first passed its simulator criteria on separate test seeds but failed on Open Bandit; we then added the misspecification-3 cells and the robustness criterion, and the third version added the per-episode basis, offset shrinkage, the COPE baseline and the grouping ablation. All of these changes were registered before the test run reported here, and their results are labelled as post-hoc where they were informed by earlier tests. The pre-registered criteria are listed in Table 5.

Metrics. We report RMSE ($\times 10^{-3}$), its ratio to the best tuned baseline in each cell with a paired bootstrap 95% interval of the RMSE difference (10,000 resamples of episodes), coverage of the nominal 95% interval, and, for the monitor, the share of episodes in which the history is split into more than one segment.

6 Results

All numbers in this section come from the one-shot test runs.

Simulated regimes. Table 2 compares O-DRIFT with the best tuned baseline in each cell; note that the best baseline differs from cell to cell, so a ratio near one means that O-DRIFT matches whichever method happens to suit that regime. Without reward drift (S0–S2) O-DRIFT is within 7% of the best baseline; the

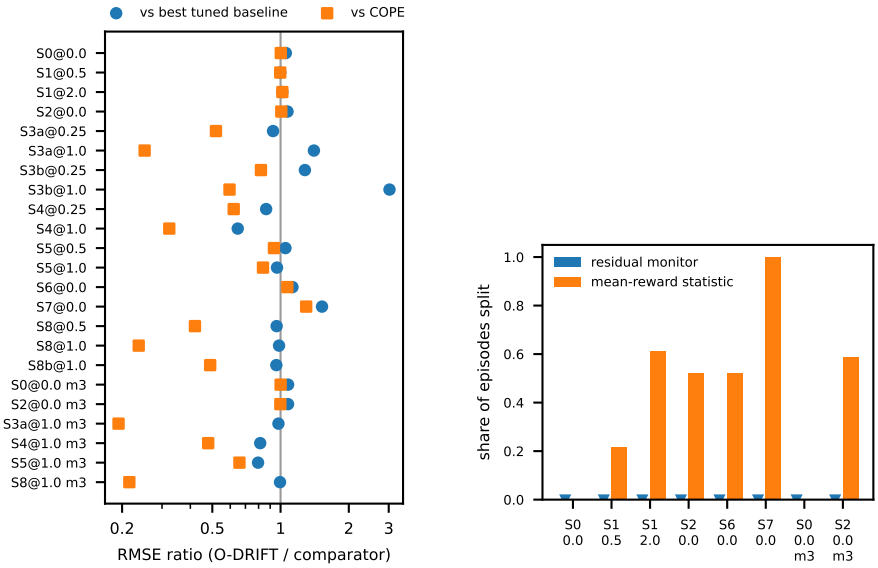


Fig. 1. Left: RMSE ratio of O-DRIFT to the best tuned baseline and to tuned COPE in each test cell (log scale). Right: share of test episodes in which the history is split into more than one segment, in regimes without reward drift, for the residual monitor and for the mean-reward statistic (same segmentation procedure, same threshold grid).

small losses in S0 and S2 are the price of estimating segment models that turn out to be unnecessary. Under abrupt and mixed reward drift it wins: by 7% in S3a at the lower intensity and by 14% and 35% in S4, with paired intervals that exclude zero. In S4 the best baseline changes from balanced DR to transported recent-window DR as the intensity grows, while O-DRIFT is best in both. The misspecification-3 cells of S3a, S4, S5 and S8 are also won or tied (ratios 0.80–0.99).

The method loses in four places. Gradual reward drift (S3b) violates the piecewise assumption and favours the time-aware DR, which models a trend (ratios 1.28 and 3.02). Under abrupt drift at the higher intensity (S3a, intensity 1) O-DRIFT loses to monitor-select (1.40); here the v1 configuration without offset shrinkage would have won (6.5 vs. 7.0), so the shrinkage tuned on development data under-corrects large drift. The same holds for S5. In S7 (support loss) the estimator correctly puts no weight on support-deficient windows, but loses accuracy against min-variance pooling (1.52). The heterogeneity-inflated interval undercovers whenever bias remains, most severely in S5 and S3b (0.22–0.63); the certified sample-split interval covers in every regime but is about 18 times wider.

Table 3. Mechanism checks on the test split. Left: RMSE ratio of O-DRIFT to tuned COPE and of the blind-grouping ablation (mean-reward statistic) to O-DRIFT (*: paired 95% interval excludes 0). Right: share of episodes whose history is split into more than one segment.

			RMSE ratio		Split rate	
Regime	Int.	Misspec.	O-DRIFT/COPE	Blind/O-DRIFT	Residual	Blind
S0	0	0.5	1.00	1.00	0.00	0.00
S1	0.5	0.5	1.00	1.07*	0.00	0.22
S1	2	0.5	1.02	1.07*	0.00	0.61
S2	0	0.5	1.01	1.13*	0.00	0.52
S3a	0.25	0.5	0.52*	1.56*	1.00	0.29
S3a	1	0.5	0.25*	1.03	1.00	0.99
S3b	0.25	0.5	0.82*	1.21*	0.64	0.02
S3b	1	0.5	0.60*	1.14*	1.00	0.82
S4	0.25	0.5	0.62*	1.25*	1.00	0.88
S4	1	0.5	0.32*	1.02	1.00	1.00
S5	0.5	0.5	0.93*	1.06*	0.99	0.72
S5	1	0.5	0.84*	1.13*	1.00	0.90
S6	0	0.5	1.07*	1.08*	0.00	0.52
S7	0	0.5	1.30*	1.02*	0.00	1.00
S8	0.5	0.5	0.42*	1.58*	1.00	0.51
S8	1	0.5	0.24*	2.24*	1.00	0.50
S8b	1	0.5	0.49*	1.23*	1.00	0.65
S0	0	3	1.00	1.00	0.00	0.00
S2	0	3	1.00	1.26*	0.00	0.59
S3a	1	3	0.19*	2.35*	1.00	0.72
S4	1	3	0.48*	1.03*	1.00	0.99
S5	1	3	0.66*	1.28*	1.00	0.72
S8	1	3	0.22*	1.01	1.00	1.00

Does the mechanism matter? Two pre-registered checks address this (Table 3 and Figure 1). First, against tuned COPE, which groups windows by mean reward, O-DRIFT has lower RMSE in all 15 reward-drift cells, with ratios from 0.19 to 0.93 and paired intervals that exclude zero, and it ties COPE in S0–S2. COPE is better in S6 and S7 (1.07 and 1.30), the two regimes with faulty propensities or lost support. The tuned COPE cluster contains all windows, so in practice COPE reduces to pooling: grouping by mean reward was not informative enough to beat it. Second, the grouping ablation keeps the whole O-DRIFT pipeline and replaces only the monitor statistic by the mean reward (its tuned form). The residual monitor never splits the history when the reward is stable (S0–S2, and S6–S7), as Proposition 1 predicts, whereas the mean-reward statistic splits 22–61% of S1 histories under context drift, 52% of S2 histories under logger switches and every S7 history. These spurious splits shrink the reference segment and cost 7–13% RMSE in S1 and S2. Under reward drift the blind grouping is never better and is worse in most cells (up to 2.35×), mainly because it misses abrupt changes: in S3a at the lower intensity it detects the change in 29% of episodes, against 100% for the monitor.

Open Bandit: where drift-corrected reuse fails. Table 4 shows the pre-registered robustness test on Open Bandit. Without reward drift O-DRIFT matches the best baseline, but under reward drift it is 3.1–7.7 times worse, and almost all of its error is bias ($+14$ to $+68\times10^{-3}$). The monitor is not the cause: it finds the change

Table 4. One-shot test on semi-synthetic Open Bandit data (200 episodes per cell). RMSE and mean error (bias) $\times 10^{-3}$; v1 is O-DRIFT without offset shrinkage; monitor-select uses the same segmentation without reuse. Ratio as in Table 2.

Regime	Int.	O-DRIFT RMSE	O-DRIFT Bias	v1 RMSE	Monitor-select RMSE	Best tuned baseline	Baseline RMSE	Ratio
S0	0	3.2	+0.4	3.2	3.3	COPE	3.1	1.03
S2	0	2.8	+0.3	2.8	3.6	COPE	2.8	1.00
S3a	0.5	15.7	+13.6	7.5	4.7	Monitor-select DR	4.7	3.33*
S3a	1	36.4	+32.2	10.0	4.7	Monitor-select DR	4.7	7.74*
S4	0.5	32.0	+28.1	21.1	13.6	Time-DR	10.2	3.14*
S4	1	73.8	+66.4	31.8	13.5	T-recent DR	10.7	6.87*
S5	1	75.4	+68.0	58.5	27.3	Reg.-DR (Liu)	20.7	3.63*

Table 5. Pre-registered criteria (protocol v3) and their one-shot outcome on the test split.

Criterion	Observed	Outcome
C1 RMSE $\leq 0.9\times$ best tuned baseline (CI excludes 0) in ≥ 2 reward-drift cells	2: S4@0.25, S4@1.0	pass
C2 RMSE $\leq 1.1\times$ best tuned baseline in S0–S2		pass
C3 split-interval coverage ≥ 0.93 in every regime with an anchor		pass
C4 S7: weight on support-deficient windows ≤ 0.05 in $\geq 90\%$ of episodes	1.000	pass
C5 no anchor $\Rightarrow$ not certified in 100% of episodes	1.000	pass
R1 no reward-drift cell (misspec. 3, Open Bandit) with ratio > 1.5	7.74	**fail**
R2 ratio ≤ 1.0 in at least half of those cells	4/9	**fail**
N1a beats tuned COPE (CI excludes 0) in $\geq$ half of reward-drift cells	15/15	pass
N1b RMSE $\leq 1.1\times$ COPE in S0–S2		pass
N2a residual monitor false-split rate ≤ 0.05 in S0–S2	0.00	pass
N2b blind grouping worse than O-DRIFT (CI excludes 0) in S1 or S2	S1@0.5, S1@2.0, S2@0.0	pass
N2c blind grouping never $\leq 0.9\times$ O-DRIFT (CI excludes 0) in a reward-drift cell		pass

in every S3a episode, and monitor-select, which uses the same segmentation without reuse, is the best method in S3a and within a factor of 1.4 of the best in S4. The failure lies in the correction. The base reward is quadratic while the reward models are linear (the basis rule chose linear features in every episode), and the injected drift is three times the standard deviation of the base logit. The fitted offsets δ_j then absorb part of the drift and part of the model error, and the corrected old segments carry a bias that the anchor cannot see. Offset shrinkage makes this worse (S3a, intensity 1: 36.4 vs. 10.0 without shrinkage). The robustness criteria (R1, R2 in Table 5) therefore fail.

Can reuse be validated on the data at hand? (post-hoc). After this test we tried, on development data only, two ways to decide per episode whether to reuse corrected segments. A gate that drops a segment whose estimate differs from the reference (reference segment plus anchor) at the 5% level removed the bias but raised variance so much that O-DRIFT lost in S4 (1.11× the best baseline) and in S4 at misspecification 3 (1.72×). A rolling-origin back-test that treats

the newest history window as a pseudo-target chose among full reuse, shrunk reuse and no reuse, but its validation error (10–130×10^{-3}) was five to ten times the errors it was meant to rank, and it was worse than the best fixed choice in every cell we tried. Neither was taken to a test split. Both failures have the cause stated in Section 3: an anchor of 1,000 records gives a DR estimate with RMSE 42×10^{-3}, and a held-out window of 4,000 records about half of that, while the reuse bias that separates good from bad decisions is around 10×10^{-3}.

7 Limitations and Conclusion

O-DRIFT rests on a simple point: for OPE under drift, the age of a window matters less than which mechanism changed it. A residual test on the conditional reward, unlike tests on raw rewards, does not mistake context or logger drift for reward drift, and in our pre-registered evaluation this made the difference between O-DRIFT and a cross-domain estimator that groups periods by mean reward in every reward-drift setting.

The limitations are as important as the result. First, drift-corrected reuse depends on the reward-drift difference lying in the model class (A4); on semi-synthetic Open Bandit data, with a misspecified model and large drift, it was heavily biased, and the estimator without reuse was the better choice. Second, data-driven checks of reuse on the anchor or on a held-out window could not repair this, because samples of the size available per period cannot detect the biases involved; we see this as an information limit, not a tuning problem, and we expect it to bind any method that validates reuse on the current period alone. Third, gradual drift violates the piecewise assumption and favours methods that model a trend. Fourth, our heterogeneity-adjusted interval undercovers when bias remains, and the certified alternative is wide. Finally, Open Bandit provides real contexts and their real order but synthetic rewards and drift; an evaluation on naturally drifting logged data remains open.

Two directions follow. One is to combine the monitor with reward models flexible enough to make (A4) plausible, so that reuse is safe when it is attempted. The other is to accept the information limit and to report, for each estimate, how much undetectable drift it would take to change a deployment decision.

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